

Exercise 3: Stored Procedures

Scenario 1: The bank needs to process monthly interest for all savings accounts.

Question: Write a stored procedure **ProcessMonthlyInterest** that calculates and updates the balance of all savings accounts by applying an interest rate of 1% to the current balance.

Answer:

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest IS
BEGIN
    UPDATE savings_accounts
    SET balance = balance * 1.01; -- apply 1% interest
    COMMIT;
END;
/
```

Scenario 2: The bank wants to implement a bonus scheme for employees based on their performance.

Question: Write a stored procedure **UpdateEmployeeBonus** that updates the salary of employees in a given department by adding a bonus percentage passed as a parameter.

Answer:

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(
    p_department IN VARCHAR2,
    p_bonus_pct IN NUMBER
) IS
BEGIN
    UPDATE employees
    SET salary = salary + (salary * p_bonus_pct / 100)
    WHERE department = p_department;

    COMMIT;
```

END;

/

Scenario 3: Customers should be able to transfer funds between their accounts.

Question: Write a stored procedure **TransferFunds** that transfers a specified amount from one account to another, checking that the source account has sufficient balance before making the transfer.

Answer:

```
CREATE OR REPLACE PROCEDURE TransferFunds(
```

```
    p_source_account IN NUMBER,
```

```
    p_target_account IN NUMBER,
```

```
    p_amount IN NUMBER
```

```
) IS
```

```
    v_balance NUMBER;
```

```
BEGIN
```

```
    -- Check source account balance
```

```
    SELECT balance INTO v_balance
```

```
    FROM accounts
```

```
    WHERE account_id = p_source_account;
```

```
    IF v_balance >= p_amount THEN
```

```
        -- Deduct from source
```

```
        UPDATE accounts
```

```
        SET balance = balance - p_amount
```

```
        WHERE account_id = p_source_account;
```

```
        -- Add to target
```

```
        UPDATE accounts
```

```
SET balance = balance + p_amount
```

```
WHERE account_id = p_target_account;
```

```
COMMIT;
```

```
ELSE
```

```
DBMS_OUTPUT.PUT_LINE('Insufficient balance in source account.');
```

```
END IF;
```

```
END;
```

```
/
```